

The Modern Informed Citizen: Understanding Trade-offs in Digital News Consumption

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ABSTRACT

In local and global contexts, the link between individual news consumption and beneficial or harmful effects is debated. Some communication theories posit that individuals must stay up-to-date with news to make civic choices like voting, volunteering, and participating in community organizing. Nonetheless, there is increasing evidence that overuse or overreliance on news can have harmful effects on mental health and lead to news avoidance. In a contested and polarized news landscape, understanding how individuals can sustain productive, healthy engagement with civic information is vital. In my work, I draw on literature in the communication and journalism fields to understand the impact of shifting media ecosystems on individuals. My studies employ methods from computational social science and human-computer interaction to analyze news content and exposure in more nuanced ways than previously available. I outline my agenda to run experiments to identify risk-benefit trade-offs at the individual level for engaging with evolving technological news systems.

CCS CONCEPTS

• Human-centered computing;

KEYWORDS

information ecosystems, news, trace data, local news, communication, journalism

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1 INTRODUCTION

My work seeks to identify and measure the types of civic information we need to be a functional society, and the conditions under which individuals make information trade-offs. I define information trade-offs as the idea that, while individuals may get benefits or even perceive it is their duty to follow current news, there may also be adverse effects from overexposure to news. In my work, I

try to understand what these trade-offs are, where we make them, and how our information ecosystems may shape our ability to manage them. I take a computational social science approach to these problems that engages deeply with the literature in the communication and journalism fields, while employing more complex computational methodologies for retrieving and analyzing data.

The “informed citizen” point of view claims that the central aim of the news media is to provide the public with key information to take actions pivotal to democracy: vote, volunteer, organize, protest. Communities have specific “information needs,” which the press helps to address [37]. Under this view, the public also have a duty to be informed about civic issues and act upon them [12]. Critiques of the informed citizens highlight that more nuanced perspectives are key to porting the “informed citizen” concept to modern life, stating that no citizen can be perfectly informed, information needs will vary between people, people should play a monitorial role, and that news resistance can be a valid form of civic participation [21, 32, 39, 40].

Communication scholars have sought to empirically identify whether news media are successful at achieving civic aims by informing and influencing individuals to take community-minded actions. Consistently, studies have shown that there is a positive correlation between the presence and readership of newspapers in an area and political turnout [16, 17, 22]. Prior work has also identified effects of news exposure on individual attitudes and civic participation, though these have been harder to causally demonstrate [10, 18, 20, 31]. A new meta-analysis of hundreds of articles on the effects of digital media on democracy found that digital media may increase political participation and information consumption, while also driving negative trends like loss of political trust and increased polarization [25].

In a digital age, media has faced new hurdles for communicating vital information with the public. Technology has impacted every stage of the journalistic production process, from how stories get sourced, written, and distributed, to who gets to tell them [13]. The state of local media in particular has been under threat, as over one fourth of local newspapers in the U.S. have shut in the last 15 years, a trend that was precipitated during the COVID-19 pandemic, leaving many communities without reliable news [2, 38]. Additionally, we have seen new threats impact the public communication process in the form of online misinformation, harassment, and trolling.

At the same time, technological advances in trace data collection and text analysis have made it possible to analyze news content and civic information shared online in novel ways. Traditional approaches to analyzing effects of news on individuals have historically relied on self-reports and surveys, but these methods are

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becoming harder to generalize across various platforms and engagement modes. De Vreese and Boomgaarden [11] produced an early example of a content analysis approach to measuring news exposure, showing that the specific content of news stories plays an important role in determining how news exposure affects individuals. Digital trace data is now available at granular levels over longer periods of time, allowing for nuanced analysis of civic information encountered online (as in [7, 27, 30]). A plethora of methods have made also it possible to label, detect, and analyze news content [15].

In my work, I answer questions about how the “informed citizen” concept is being reshaped in digital spaces through 1) an understanding of how information tradeoffs are molded by digital media and civic information in non-news spaces, and 2) linking analyses of observational and trace data together with survey data and contextual interviews.

2 CONTRIBUTIONS TO DATE

My prior work has used qualitative and quantitative approaches to identify how people select news headlines to engage with, the types of news content that people have access to, and how people manage contested information landscapes.

In the first study of my PhD, I explored how people engage with news headlines geared to maximize engagement. I curated, analyzed, and released a large-scale dataset of more than 30,000 A/B tested news headlines from the digital publisher Upworthy between 2013–2015 [26]. I used the dataset to validate the psychological theory of the *information gap* in the field [24] through a meta-analysis. In a journalistic context, the information gap is the tactical withholding of information from a headline to entice a reader to click on an article. I modeled this concept through creating and validating a novel automated method for tagging headlines with linguistic concreteness. From analyzing the data, I empirically demonstrated that linguistic concreteness maximizes clickthrough rates at middling levels, rather than when headlines are too vague or too concrete. This will be the first time the theory has been linked to online data, and ties lab-derived psychological insights to the ways humans dedicate attention to news online [5].

Another project I led sought to understand the societal factors that impact local media coverage and engagement at scale. Understanding who has access to which types of local media was particularly relevant during the COVID-19 pandemic, as people needed information about how to protect themselves and their communities. These dynamics are well-studied for national media, yet remain challenging to analyze at scale for local media due to the distributed nature of outlets. Since comprehensive datasets of local media stories are scarce, I used computational methods to curate the largest known dataset of local news sources in the U.S. To this end, I scraped known local news outlets on Facebook for their “Related Pages,” which allowed me to aggregate 10,258 local news outlets, their Facebook and Twitter handles, and their posts over a multi-year period. Using topic modeling and multilevel modeling, I found that coverage of COVID-19 varied significantly based on the partisanship of the region and the estimated “localness” of an outlet. In particular, we identified that more local outlets covered case counts proportionally more, while more national outlets tended to cover the economic effects of COVID-19 proportionally more. This work

is the largest analysis of local news outlets to date, and represents the first time the impact of “localness” on topic coverage has been demonstrated. Our paper on this topic was published in ICWSM 2022 [4]. This is an example of how computational techniques can allow for the novel collection and analysis of data, thus expanding and adding nuance to communication literature.

Finally, I investigated how essential workers navigated a contested information landscape during the early months of COVID-19 [3]. In a series of interviews, I asked essential workers to recount how they sought out information about the pandemic, the news and information sources they turned to, and how they evaluated them. Participants highlighted issues around staying up-to-date with changing workplace protocols, assessing personal daily risk, and keeping up with news coverage. In particular, we found that the need to stay continually informed in a crisis environment took a mental toll on participants, many of whom developed coping strategies to manage information intake. These findings contributed to an understanding of 1) how our current contested media environment impacts those who keep fundamental services running in times of crisis and 2) how workplaces can help manage the mental strain for their employees in future crises.

3 EXPLORING INFORMEDNESS-SAFETY TRADE-OFFS FROM LOCAL SOCIAL MEDIA PLATFORM USE

Scholars hypothesize that social media and local groups might be “filling the void” that local news outlets leave behind as they are shuttering across the country [1, 38], yet the effects of this shift have not been empirically demonstrated. In recent years, there has been an increase of people getting local information from non-traditional sources, and particularly from online sources. For example, in a Pew survey, 59% of respondents stated they access local information from online forums or discussion groups [29]. Prior communication literature has found robust correlations between local news use, individual pro-community attitudes, and civic participation [17, 33, 35]. However, as people turn more and more to informal local online spaces to replace struggling local news outlets, we do not know if these effects will hold. With people’s attention shifting from legacy local media outlets to online local spaces, there is a need to understand how this may impact people’s attitudes and behaviors.

Current work. In my current work, I have instrumented computational methods that alter participant information exposure and use survey methods and community data to measure the causal impact of the intervention. I conducted a field experiment where heavy Facebook users in four U.S. cities were randomized to follow either a *Local News Page*, a *Social Local Online Group*, and a *Control* set of non-local hobby groups. I created a framework that allows the assessment of change in individual-level community attitudes (e.g. community involvement and civic attitudes) pre- and post-treatment. Through pre- and post-experiment surveys, I evaluated whether participant attitudes to their local communities changed based on the types of local information they were exposed to on Facebook. Though participant community attitudes were not causally impacted by exposure differences, they had significantly

different perceptions of the quality of information sources. In particular, participants trusted and were more interested in *Local News Pages* than *Social Local Online Groups*, indicating that interventions may be necessary to preserve civic information as people increasingly turn to social media for local news. This work was presented at the Computation + Journalism 2022 symposium and has been submitted to the International Communication Association conference 2023 [6].

Future Work. Next steps we plan to take include extending the work to other platforms, working more closely with the participants, and running the study for a longer period of time. While Facebook groups has been one popular form of local online social media, there are others which have raised specific concerns about trust in communities. In our next study, we will focus on individual trade-offs that result from subscribing to a local news media newsletter relative to a local safety app. We will focus on the mobile app Citizen, which brands itself as a tool to help people stay “safe and informed” in their neighborhoods [9]. On the one hand, Citizen may be contributing to the same benefits as local media, through enabling people to be more informed about local events. However, others have critiqued the app as pushing surveillance and fear to capitalize on its business model [19]. We will look at the media effects of these local information sources from three angles: impact on self-perceived local informedness, sense of safety, community satisfaction, and civic attitudes. In particular, we will seek to quantify if there are trade-offs between self-perceived local informedness and civic attitudes, and personal sense of safety and satisfaction. Although this work is still being conceived, we are hoping to work with local journalist entities, such as TheCity in New York City, to help participant recruitment. Once completed, this work will help us to understand how shifting local ecosystems are likely to impact populations.

4 IDENTIFYING TRADE-OFFS IN BALANCING PERSONAL DIGITAL INFORMATION WELLBEING

Studies on Covid-19 information seeking behavior precipitated recent findings on potential adverse mental health effects of following news too closely. Scholars that investigated media use during the early months of COVID-19 found a consistent trend that high exposure to news during this crisis period could lead to information overload and take a toll on people’s mental health [8, 14, 23]. These studies dovetail into an emerging literature on the phenomenon of a subset of people for whom perceived information overload leads to active avoidance of news [28, 34, 36]. I also witnessed these mental health effects and coping behaviors in our interviews with essential workers, who struggled acutely when managing a contested information environment while facing work pressure [3].

Current work. I am working with a medium-sized news aggregator to investigate how personal news tracking tools may allow people to reflect on their own media habits. The goal of this study is to understand whether people’s expectations of the news they consume aligns with reality, and what their response is when shown their aggregated news behaviors. This study will allow us to explore whether people perceive news tracking to be a valuable avenue for

reflecting and managing news choices. While a few news tracking products exist, these applications have largely not been examined in the academic literature. Our mixed-method study consists of three parts: 1) collecting trace data about people’s usage of the news aggregator, 2) building a prototype to reflect the news habits back to users, and 3) interviewing people about their self-perceived news habits before and after seeing the prototype with their own data. Currently, we have obtained IRB approval, collected pilot data from the news aggregator, built the prototype, and will soon be beginning pilot interviews. With this work, we hope to illuminate how seeing their data using a self-tracking tool may lead people to reflect on their news habits. This study is a novel approach to combining trace data with qualitative interviews in the news space, and builds on communication concepts using computational methods.

Future work. For my dissertation, I am planning a set of projects to identify trade-offs between closely following the news and the toll on people’s mental health. These will continue to be in collaboration with the news aggregator. I will use survey-based techniques and analysis of content data, to understand how close engagement with news impacts people’s information wellbeing. The key to this analysis will be to set up survey-based measures for 1) expected benefits of increased news exposure (such as perceptions of informedness, civic attitudes, etc.) and 2) expected mental health risks of increased news exposure (such as difficulty sleeping, higher stress levels). I will then test if, based on the survey results, there is a trade-off between the expected benefits of news consumption and the toll on mental health. Additionally, I will analyze if the types of news these users are engaging most with have an impact on the trade-offs. As an extension of this project, I am planning to construct an experiment that will show if this trade-off can be mediated by environmental factors. Through working with this news aggregator, we will set up a field experiment, with 1) a group whose volume of news notifications is randomly reduced by 5%, 2) a group whose volume of political news notifications is reduced by 5%, 3) a group whose volume of entertainment news notifications is reduced by 5%, and 4) a control group whose notification volume is not reduced. We will then run a final survey among the experimental groups of users to understand how their information wellbeing was impacted by the changes. These studies will provide large-scale, causal evidence of the existence of information trade-offs between the “informed citizen” and human ability to balance information intake.

5 CONCLUSION AND NEXT STEPS

While “informed citizen” theories posit that staying up-to-date on current events is important for civic engagement, there is evidence that overuse of news can have harmful effects on individual mental health. A wealth of literature in the communication space has developed survey-based social science instruments to measure these concepts using legacy media concepts. However, as media shifts and evolves through technology, new studies and methods are required to understand how these ecosystem changes impact people. Additionally, new techniques using and analyzing observational data can draw more accurate conclusions about news exposure and impact. I hope that my work will continue to straddle both

communication and human-computer interaction literature, to develop methods for finding risk-benefit trade-offs at the individual level. Findings in this domain will help the more mindful design of platforms, contribute towards a deeper understanding of how the media impacts people's attitudes and actions, and inform the design of effective solutions for modern media problems.

Currently, I am a fourth-year PhD candidate in the Information Science department at Cornell Tech. I am now turning my attention towards my dissertation, and I am planning for the two to three studies I am likely to undertake in the last years of the PhD. In the longer term, I am considering research positions in academia and industry.

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